Interpretation: features

5.11 Colour-coded archaeological interpretation plans are provided in Figures 5 and 8.

General comments

5.12 Except where stated otherwise in the text below, positive magnetic anomalies are taken to reflect relatively high magnetic susceptibility materials, typically sediments in cut archaeological features (such as furrows, ditches or pits) whose magnetic susceptibility has been enhanced by decomposed organic matter or by burning.

5.13 Series of parallel, weak, positive magnetic anomalies which almost certainly reflect former ridge and furrow cultivation have been detected across Phase 2: Areas 1A, 2B, 7B; Phase 3: 1, 2A, 2B, 3A, 5.

5.14 Small, discrete dipolar magnetic anomalies have been detected in all of the survey areas. These almost certainly reflect items of near-surface ferrous and/or fired debris, such as horseshoes and brick fragments, and in most cases have little or no archaeological significance. A sample of these is shown on the geophysical interpretation plans, however, they have been omitted from the archaeological interpretation plans and the following discussion.

5.15 In the text below the surveys are described with Phase 2 followed by Phase 3. Potential archaeological features have been assigned feature numbers [Fx], which are also indicated on the archaeological interpretation drawings.

Phase 2: Areas 1A and 1B (Figures 6-8)

5.16 Two positive magnetic anomalies [F1, F2] have been detected at the southwestern end of Area 1A. These may reflect soil-filled features, possibly pits or ditches.

5.17 A farm track situated next to and parallel with the northeastern field boundary of Area 1A has been detected as a series of intense dipolar magnetic anomalies. A farm gate has also been detected as a dipolar magnetic anomaly on the northern boundary of this survey area.

5.18 A group of dipolar magnetic anomalies [F3] at the northwestern end of Area 1A almost certainly reflects a concentration of ferrous or fired materials.

5.19 Also detected:
Traces of ridge and furrow cultivation [F4] oriented northwest-southeast in Area 1A.
The modern plough regime has been detected in Phase 2: Area 1B. This is probably on the same alignment as the medieval ridge and furrow however it is not possible to differentiate the two.

Phase 2: Areas 2A, 2B and 4; Phase: 3 Areas 1, 2A and 2B (Figures 3-5)

5.20 An extensive complex of predominantly linear, rectilinear and sub-circular positive magnetic anomalies has been identified across these areas. The vast majority of these anomalies reflect soil-filled features, which define roads, trackways and ditches while the smaller anomalies reflect pits and gullies (a sample of these features is discussed in the following text). Together these form a palimpsest of features of several phases over six hectares of survey but clearly extending beyond those surveys in all directions.

5.21 Two double-ditched trackways or roads [F6, F7] were detected as strong positive magnetic anomalies. The first [F6], oriented north-south, crosses Phase 3: Areas 2A and 2B, and possibly continues into Phase2: Area 4, though it may pass just to the east of this area. The second road [F7] joins the first at its widest point (nearly 10m across) in Phase 3: Area 2B, to form a T-junction. It then winds northeast through Phase 2: Areas 2A and 2B; the irregular nature of [F7] may indicate that it was routed around another feature or structure to the north of the survey area.

5.22 A large rectilinear enclosure [F8], approximately 40m by 60m and oriented on a northwest-southeast alignment, frames the T-junction of the two roads. This suggests that the roads are not contemporary with this enclosure ditch.

5.23 Two further rectilinear enclosures [F9] have been identified in Phase 2: Area 4 to the south of [F8]. These contain many positive magnetic anomalies, almost certainly reflecting internal ditches, pits and gullies. Two large probable pits [F10, F11] are present at the edge of these enclosures, one at the interchange of three ditches.

5.24 A possible boundary or enclosure ditch [F12] at the northwestern extent of Phase 3: Area 2A appears to define the limit of settlement there.

5.25 Two narrow positive magnetic anomalies [F13], probably soil-filled ditches, form a T-junction with another ditch and a large probable pit feature [F14] in the centre of Phase 2: Area 2A. These represent a different phase of activity to the track [F7] which they cut or are cut by. A further ditch [F15] orientated northwest-southeast appears to be associated with these features.

5.26 Several positive magnetic linear anomalies [F25] have been detected in Phase 3: Area 1, probably reflecting soil-filled ditch and pit features.

5.27 Also detected:
Traces of medieval ridge and furrow cultivation [F16, F17] oriented northeast-southwest and northwest-southeast. The ridge and furrow anomalies are of a higher intensity here, due to the large amounts of relatively high magnetic susceptibility material which has been ploughed up from the underlying features.

Traces of ridge and furrow cultivation [F26] orientated east-west in Phase 3: Area 1.

Phase 2: Areas 5A and 5E (Figures 3-5)

5.28 A number of positive magnetic anomalies [F18] have been detected across these areas, which may reflect soil-filled features, possibly pits, gullies and ditches.

5.29 A chain of dipolar magnetic anomalies orientated north-south on the western edge of the survey area almost certainly reflects a modern service.

5.30 The magnetic nature of the field boundaries, the railway to the north and the main A-road to the south have all been detected as large dipolar magnetic anomalies.

Phase 2: Areas 6A and 6B (Figures 6-8)

5.31 Three circular positive magnetic anomalies [F19, F20 and F21] have been detected which may be associated with prehistoric roundhouses. Each anomaly is fairly uniform in both size (8-10m diameter) and strength and each has a positive or dipolar magnetic anomaly to its centre, which may reflect a central hearth within the building.

5.32 Several other positive magnetic anomalies [F22] have been detected in Phase 2: Area 6B which may reflect soil-filled pits or ditches.

Phase 2: Areas 7A and 7B (Figures 6-8)

5.33 Three positive magnetic anomalies [F23] have been detected in this area. These may reflect soil-filled features.

5.34 Large concentrations of dipolar magnetic anomalies have been detected at the entrances to the field in which survey 7A took place. These reflect a water trough and plough which were present at the northern and central entrances respectively. Around these items were concentrations of ferrous and fired materials such as brick, which had been used to create a firm track.

5.35 Also detected:
Traces of possible ridge and furrow cultivation [F24] orientated northwest-southeast in Phase 2: Area 7B.

Phase 3: Areas 3A and 3B (Figures 6-8)

5.36 A complex of positive, negative and dipolar magnetic anomalies has been detected across the western extent of Phase 3: Area 3A. These could reflect an area of former occupation that may be part of, or related to, the large area of settlement detected to the west.

5.37 A linear negative magnetic anomaly [F27] oriented northwest-southeast has been detected. This may reflect the metalled surface of a former road or trackway. This probable trackway diverges around enclosure [F28]. Internal structures and pits have been detected within this enclosure.

5.38 A second enclosure [F29] has been detected to the east of [F27]. This follows a similar pattern of a rectilinear negative anomaly, possibly reflecting a walled

enclosure, an associated metalled track, and internal positive and dipolar magnetic anomalies probably reflecting structures, pits and gullies.

5.39 Further positive magnetic anomalies [F30] have been detected between enclosures [F28] and [F29]. These almost certainly reflect further enclosures, trackways, pits and possibly structural features.

5.40 A large area of dipolar magnetic anomalies towards the middle of the survey area reflects the large amounts of ferrous and fired debris which surround a large cattle feeder.

5.41 Further small, weak positive magnetic anomalies [F31] have been detected in the eastern part of Phase 3: Area 3A and Area 3B. These may reflect the truncated remains of further pits and gullies.

5.42 A pair of perpendicular positive magnetic anomalies in Phase 3: Area 3A may reflect soil-filled features, possibly former field boundaries.

5.43 A former animal water-hole, now backfilled, has been detected as large dipolar magnetic anomalies.

5.44 The existing wire fence boundary which traverses part of Phase 3: Area 3A on a northwest-southeast alignment has been detected as a chain of small dipolar magnetic anomalies.

5.45 Also detected:
Traces of ridge and furrow cultivation [F32, F33] both oriented northeast-southwest. The westernmost ridge and furrow anomalies are of a higher intensity due to the large amounts of magnetic material that the plough has brought up from the features beneath.

Phase 3: Area 4 (Figures 3-5)

5.46 An area of probable occupation [F34] has been identified within this land parcel. It comprises a series of strong linear and discrete positive magnetic anomalies which almost certainly form a complex of ditches and enclosures. Intense dipolar magnetic anomalies detected in the southern part of this complex may reflect hearths or kilns [F35], indicating that some industrial activity was taking place. The complex appears to extend to the south and north, however, it will have been severely truncated at its northern end by the Wilts and Berks Canal [F36] (now in-filled in this section).

5.47 To the west of the main complex are several negative magnetic linear anomalies [F37] which could reflect possible stone structures. These again seem to form walled enclosures.

5.48 The Wilts and Berks Canal has been detected as a series of intense dipolar magnetic anomalies on a northeast-southwest alignment. Three large dipolar magnetic anomalies and several concentrations of discrete dipolar magnetic

anomalies have been detected adjacent to the canal, which may all indicate features associated with the waterway.

5.49 Several other dipolar magnetic anomalies have been detected in this survey area, which reflect a variety of modern farm items such as a barn, wire fences, metal gates and a water trough.

Phase 3: Area 5 (Figures 6-8)

5.50 A weak, discontinuous, positive magnetic anomaly [F38] with curved corners has been detected which may reflect an enclosure of prehistoric date. It is approximately 40m square and contains two further positive magnetic curvilinear anomalies: a possible ditch or gully [F39] and a possible a ring-ditch or internal structure such as a roundhouse [F40].

5.51 A further two positive magnetic linear anomalies [F41] have been identified to the northwest of the square enclosure which may reflect associated land boundaries for activities such as stock raising.

5.52 Several other positive magnetic anomalies are present within the survey area, which could reflect pits and gullies associated with the possible enclosure.

5.53 Also detected:
Traces of ridge and furrow cultivation [F42] oriented northwest-southeast.
Probable land drains, identified as a complex of negative magnetic anomalies.

6. Conclusions

6.1 Thirty-seven hectares of geomagnetic survey were undertaken on land to the east of Swindon.

6.2 Soil-filled features including ditches, enclosures, pits and gullies have been identified across the survey areas; these can be roughly divided into five foci.

6.3 A large multi-phased area of occupation possibly of Romano-British date has been identified around South Marston Farm, (survey areas Phase 1: B1; Phase 2: 2A, 2B, 4; Phase 3: 1, 2A, 2B). This settlement extends beyond those areas already surveyed.

6.4 Trackways and enclosures have also been identified in Phase 3: Area 3A to the east of South Marston Farm. These may be related to the much larger area of occupation to the west.

6.5 An area of occupation identified during the Phase 1 surveys (Area E1) south of Marston Farm has been found to extend beyond the original survey. Possible stone walls and ditch features have been identified across the survey extension Phase 3: Area 4.

6.6 A probable prehistoric enclosure has been identified in Phase 3: Area 5.

6.7 Three possible prehistoric roundhouses have been identified in Phase 2: Areas 6A and 6B.

6.8 Traces of probable medieval ridge and furrow cultivation have been identified in Phase 2: Areas1A, 2B, 7B; Phase 3: 1, 2A, 2B, 3A, 5.

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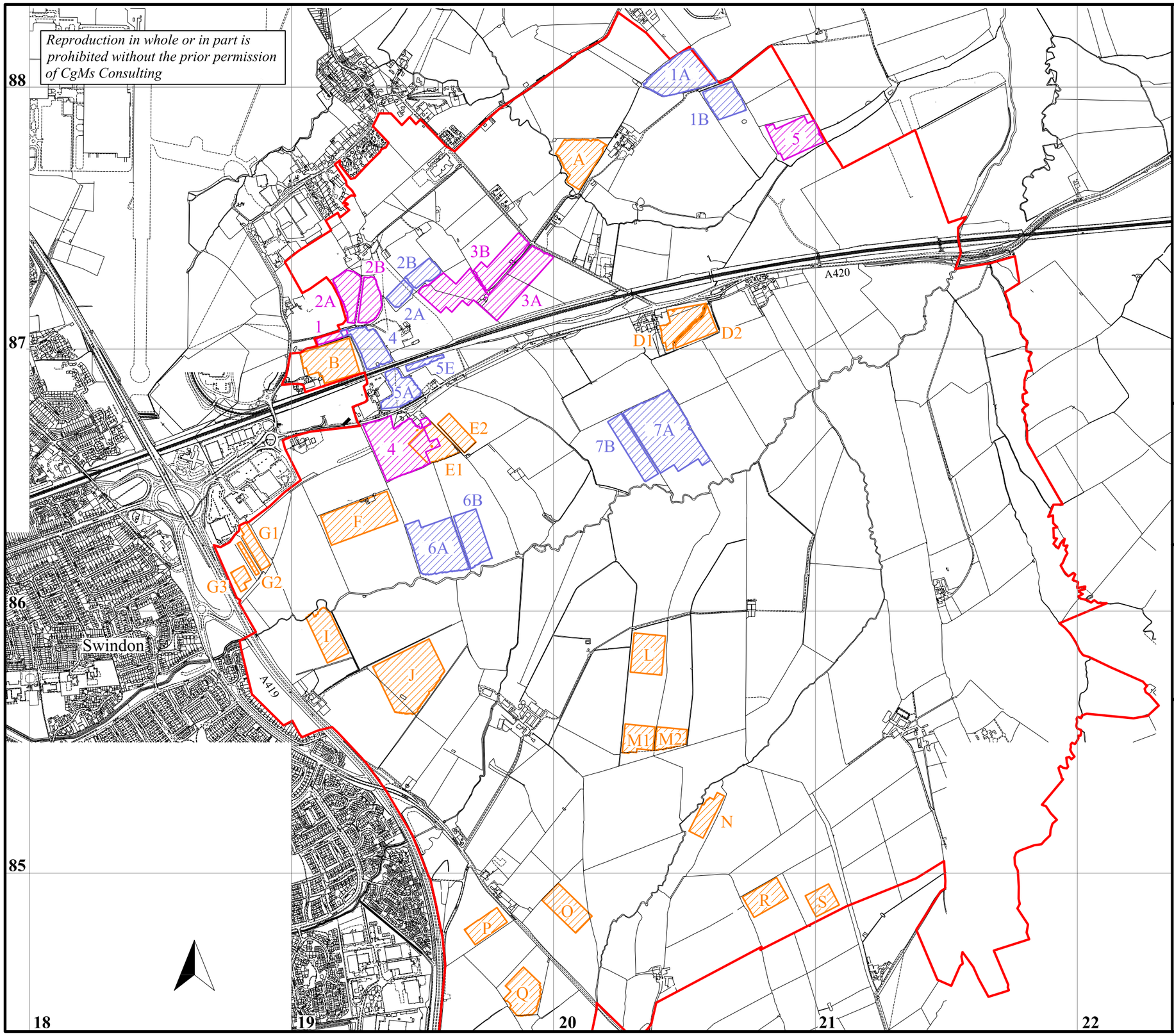

Archaeological Services
University of Durham

Land east of Swindon, Wiltshire

geophysical surveys - phases 2 and 3

Report 1746

Figure 1

Location of survey areas, phases 1-3

on behalf of
CgMs Consulting

0 500m
scale 1:15 000 - for A3 plot

site boundary
phase 1 survey area
phase 2 survey area
phase 3 survey area